Supplementary Figure 6-4. Extended-range in vitro oxygen-ploidy response

O2 = 0-20% at 0.1% intervals. Values above 5% are post-fit model extrapolations.

A. Separate-fit in vitro oxygen-ploidy response classes

500 fitted seeds; black, class median; dashed line, fitted O2 boundary

Complex nonmonotone (n=111)

Inverted U (n=152)

Monotone increasing (n=45)

U-shaped (n=35)

Approximately flat (n=103)

Increase then plateau (n=54)

Decrease then plateau (n=0)

Monotone decreasing (n=0)

Smoothed dominant mean ploidy

Fixed oxygen (%)

B. Oxygen-missegregation-ploidy response surface

q10 ensemble; purple, weak-gap region; black and gray, fitted p_miss,eff median and 10-90%

C01 C02 C03

Effective p_{miss}

Fixed oxygen (%)

Median dominant mean ploidy (log colors)

6.0 4.0 3.0 2.0 1.5 1.0

C. Effective missegregation required for target ploidy

Gray, no stable unique inverse; purple, multiple inverse solutions

C01 C02 C03

Target dominant mean ploidy

Fixed oxygen (%)

Median required p_miss,eff (log colors)

0.500 0.100 0.050 0.010 0.005

Reference curves

- p_miss,eff = 0.01
- p_miss,eff = 0.10
- p_miss,eff = 0.20
- p_miss,eff = 0.30
- Mean across 496 fixed p_miss,eff values

D. Ploidy consensus within weak-gap regions

C01 C02 C03

Effective p_{miss}

Fixed oxygen (%)

Across retained seed endpoints

- >=90% low (<=2)
- >=90% intermediate (>2 and <4)
- >=90% high (>=4)
- <90% three-class agreement

E. Local rank-1 regime-switch sensitivity

C01 C02 C03

Effective p_{miss}

Fixed oxygen (%)

Endpoints changing class at any 4-neighbor cell

100% 75% 50% 25% 0%

F. Across-fit dominant-mean-ploidy spread

C01 C02 C03

Effective p_{miss}

Fixed oxygen (%)

90th-10th percentile spread (mean-ploidy units)

0.500 0.100 0.010 0.000